Domain-Specific LLM Adaptation: Transformer Pipeline for Medical Question Answering

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ABSTRACT

The rapid evolution of Large Language Models (LLMs) has facilitated great modifications to natural language understanding in different domains. Although most modern LLMs are trained on rather generalistic data, their use in highly specialized and precise fields like medicine can be limited by factors like factual accuracy and domain language. This paper aims to analyze the effectiveness of fine-tuning the pretrained LLM FLAN-T5 in the medical question-answering domain. The two popular medical datasets PubMedQA and MedMCQA were used to fine-tune the model for enhanced comprehension of biomedical terminology, clinical situations, and evidence-based reasoning.

The training pipeline was built on the HuggingFace Transformers framework with meticulous attention to preprocessing, optimization strategies, and evaluation protocols. The performance evaluation was thorough, including multiple metrics like Accuracy, BLEU, ROUGE, BERTScore, and USEScore to determine the quality, fluency, and semantic alignment of the ascertainment. Results exhibit major improvements in domain relevance and understanding of questions post fine-tuning, evidenced in datasets that require scientific grounding. The paper thus brings forth the potential role of task-specific tuning in enhancing LLM performance in sensitive knowledge domains and towards scalable, accurate, and reliable language models for real-world medical applications.

INTRODUCTION

Large Language Models (LLMs) stand tall in the field of Natural Language Processing (NLP) with their sheer strength for machine translation, summarization, answering queries, and so on. Models such as GPT, BERT, FLAN-T5, etc., have hard-earned their reputation of performing with competence in zero- and few-shot scenarios on a myriad of benchmarks, as they were pre-trained on large volumes of data pertinent to the general domain. [1] On the contrary, the effectiveness of such models is bound to wane in highly specialized domains such as medicine, where a level of expert knowledge, precision, and contextual reasoning must apply.

In recent years, the healthcare industry have churned out an enormous volume of unstructured text data concerning research articles, clinical notes, patient queries, etc. If LLMs can be harnessed to precisely understand and act upon such content, much potential could be anticipated in clinical decision-support systems, patient education, and medical research. Nonetheless, the generalized pre-training restricts these models from acquiring deep understanding, domain-specific reasoning, and limited accuracy in complex question answering skills. With the intention of bridging this gap, the research focuses on fine-tuning a pretrained LLM, FLAN-T5, on medical datasets so that it may perform better on domain-specific question-answering tasks. Being an instruction-tuned model, FLAN-T5 is quite powerful and generalizes well to unseen tasks. By tailoring it to the medical domain, the goal is to improve the model's ability to answer medical questions with respect to accuracy, coherence, and semantic alignment in clinical knowledge.

To aid that, two famous datasets were selected: PubMedQA, an open-ended biomedical research-question dataset [3]; and MedMCQA, a multiple-choice medical exam dataset. [4] Offering different question and answer formats, these datasets provide a wider backdrop for gauging model performance. The fine-tuning was carried out using the HuggingFace Transformers library and evaluated on various parameters [7] using Accuracy, BLEU [8], ROUGE [9], BERTScore [10], and USEScore metrics [11].

This work adds to the growing body of research in LLM domain adaptation and offers some insights into how instruction-tuned models such as FLAN-T5 can be further extended to critical domains. The findings indicate that with fine-tuning targeted specifically towards certain tasks, even general-purpose models can achieve some promising results in medical question answering.

RELATED WORK

Large-language models such as BERT, GPT, and T5 have revolutionized the domain of natural language understanding and generation. The bidirectional contextual embeddings introduced by BERT offered an immense leverage in many NLP tasks by understanding context from both directions-left and right. GPT models, on the other hand, would operate in an autoregressive style, which is optimized for generating long and coherent text. While T5's encoder-decoder architecture frames all tasks as text-to-text problems, offering versatility across diverse NLP applications. [1]

The adaptation of such generic LLMs to medicine is crucial for this domain whose vocabulary and reasoning are far removed from what generic LLMs can handle. Consequently, domain-specific models like BioBERT [5] and ClinicalBERT[6], through their pre-training on a large corpus of biomedical literature and clinical notes, respectively, demonstrated improved performance for named entity recognition and medical question answering.

In medical QA, datasets such as PubMedQA pose open-ended research questions in biomedical areas that need factually correct and contextually coherent answers, supported by the given scientific abstracts. [3] In contrast, multiple-choice medical QA datasets such as MedMCQA[4] are important for checking whether a model is endowed with medical knowledge in an educational or assessment environment and require reasoning skills with clinical concepts with a high degree of precision. Recently fine-tuned LLMs on these datasets have shown promising accuracy but face problems such as hallucination[12] and lack of interpretability.

By contrast, instruction tuning, as demonstrated in FLAN-T5, further made LLMs highly adaptable, as they could now generalize across tasks by following natural language instructions with very little task-specific data [13]. Nevertheless, very few works evaluated instruction-fine-tuned LLMs on multiple, heterogeneous medical QA datasets.

This work aims to extend the field by fine-tuning FLAN-T5 on PubMedQA and MedMCQA, which combine open-ended and multiple-choice QA tasks. This allows for a systematic evaluation of instruction-tuned LLMs on specialized medical domains, giving insights into cross-dataset generalization, task robustness, and factual consistency.

METHODOLOGY

The framework outlined here describes the steps taken to create a domain-oriented medical question-answering method built on the FLAN-T5 model type. The entire framework spans from picking an appropriate pretrained language model, while engaging in dataset preparation, training setups, and evaluating model performance using traditional and semantic evaluation criteria.

MODEL SELECTION: WHY FLAN-T5?

For this project, I decided to use Google's FLAN-T5 (Fine-tuned Language Net T5), a potent instruction-tuned variant of the T5 (Text-To-Text Transfer Transformer) architecture, for this project. FLAN-T5 is well suited for this domain-specific task because it is especially good at multi-task generalization and performs well on tasks that require question answering, summarizing, and following instructions. [1]

By reformulating all tasks into a single text-to-text format, FLAN-T5 improves on the original T5 model, which was trained on a sizable corpus (also known as the Colossal Clean Crawled Corpus, or C4). Using carefully selected instruction-following datasets, FLAN improves T5's generalization and alignment with user intent. [2] The FLAN-T5-Base variant was selected due to its ability to balance performance and computational efficiency, allowing for fine-tuning on low-end hardware while still producing significant results in specialized domains like biomedicine.

DATASET PREPARATION

For this project I used two domain-specific datasets: MedMCQA and PubMedQA.

- **PubMedQA:** A dataset of biomedical question-answering, taken from PubMed articles is called PubMedQA.[3] It includes contextual abstracts, long-form expert-generated responses, and research questions (yes/no/maybe). I chose the first 10% of the training split for quick prototyping and used the "pqa_labeled" version for this project. To guarantee high-quality supervised learning, only samples with non-null long answers were kept.

  To prepare it for FLAN-T5, the data was preprocessed into a sequence-to-sequence format:

  - The input_text was structured as:

    question: <question_text> context: <abstract_text>

  - The target_text was the long-form expert answer.

- **MedMCQA**: A sizable multiple-choice QA dataset called MedMCQA was created specifically for preparing for medical exams.[4] One question, four possible answers (opa, opb, opc, and opd), and the appropriate option index (cop) make up each instance. For manageability, I utilized 10% of the training split.

  Every question was restructured to resemble an instruction:

  - input_text:

    question: <question> options: A. <opa> B. <opb> C. <opc> D. <opd>

  - target_text: The correct option text, mapped from cop.

FINE-TUNING SETUP

Tokenization, model training, and checkpoint saving were all handled by using the HuggingFace Transformers library. [7] By providing pre-built architectures and training resources for a variety of Transformer-based models, HuggingFace streamlines model workflows.

- **Tokenization:** I loaded the FLAN-T5 tokenizer using HuggingFace's AutoTokenizer. This takes care of truncation, padding, and tokenization of subwords. Outputs (targets) were truncated at 128 tokens, and inputs were truncated to a maximum length of 512 tokens.
- **Model loading:** I loaded the FLAN-T5-base model using T5ForConditionalGeneration. With support for autoregressive decoding and encoder-decoder attention, this architecture is especially made for generation tasks.
- **Training:** HuggingFace's Seq2SeqTrainer and Seq2SeqTrainingArguments were used for training. Among the crucial configurations were:
  - batch_size: 4 (per device)
  - learning_rate: 3e-4
  - epochs: 2
  - weight_decay: 0.01
  - predict_with_generate: Enabled (not just for classification logits, but also for evaluating outputs using generation)

Two models were fine-tuned separately:

- One on the PubMedQA dataset and saved as ./flan-t5-pubmedqa1
- One on the MedMCQA dataset and saved as ./flan-t5-medmcqa1

Models and tokenizers were saved locally after training for inference and evaluation.

EVALUATION METRICS

I employed a variety of automatic evaluation metrics appropriate for each task's requirements to evaluate the performance of the optimized models:

- **Accuracy**

  Used for MedMCQA (MCQ task). Calculates the percentage of predictions in which the ground-truth answer is accurately identified by the model. Simple and suitable for tasks involving the answer to objective, closed-ended questions.

- **BLEU Score (Bilingual Evaluation Understudy Score)**

  Used for PubMedQA. Calculates the n-gram overlap between the reference and predicted responses. Particularly helpful for assessing long-form generation. While BLEU can penalize legitimate paraphrases, it also captures fluency and structure. [8]

- **ROUGE Score (Recall-Oriented Understudy for Gisting Evaluation)**

  Used for PubMedQA. Focuses on recall, capturing how much of the reference answer is covered by the generated output. ROUGE-1 and ROUGE-L were primarily used to assess unigram overlap and longest common subsequence. [9]

- **Exact Match & F1-Score (EM)**

  These proxy metrics are frequently used in QA to detect hallucinations. Even if the content is grammatically correct, a low EM/F1 may suggest that the model generates factually inaccurate or hallucinogenic content. Because it contains structured reference answers for comparison, it is used on PubMedQA.

- **BERTScore**

  Compares the semantic similarity between reference and prediction using pre-trained BERT embeddings. More lenient than BLEU or ROUGE when it comes to paraphrased responses. Mostly utilized on PubMedQA. [10]

- **USEScore (Universal Sentence Encoder Similarity)**

  Calculates the cosine similarity between the generated and reference answers' sentence embeddings. Provides a high-level perspective on semantic overlap. Used on PubMedQA, specifically to assess the coherence and retention of meaning in lengthy responses. [11]

These assessment techniques guarantee that the output's syntactic (BLEU, ROUGE) and semantic (BERTScore, USEScore) components are measured. Semantic metrics are especially crucial for PubMedQA, which includes intricate, verbose responses. On the other hand, MedMCQA's deterministic multiple-choice format means that accuracy alone offers enough insight.

RESULTS AND ANALYSIS

The base and finr-tuned FLAN-T5 models are evaluated quantitatively and qualitatively on two benchmark medical QA datasets, PubMedQA and MedMCQA, in this section. Reported multiple metrics and include relevant visualizations and examples to support our analysis.

QUANTITATIVE EVALUATION

Used two medical QA datasets, PubMedQA and MedMCQA, to compare the base and fine-tuned FLAN-T5 models in order to assess their performance.

Conducted a thorough assessment of PubMedQA utilizing a variety of metrics. Since MedMCQA is a multiple-choice QA dataset, only assessed Accuracy.

Metric	Base FLAN-T5	Fine-Tuned FLAN-T5
ROUGE-1	17.53%	26.21%
ROUGE-2	6.45%	9.43%
ROUGE-L	14.37%	20.21%
BLEU	0.44%	2.32%
F1 Score	0.1380	0.2048
Exact Match (EM)	0.00	0.00
BERTScore F1	0.8462	0.8783
USEScore	0.3191	0.4421

Table 1: Performance Comparison on PubMedQA.

Metric	Base FLAN-T5	Fine-Tuned FLAN-T5
Accuracy	24 %	67 %

Table 2: Performance Comparison on MedMCQA.

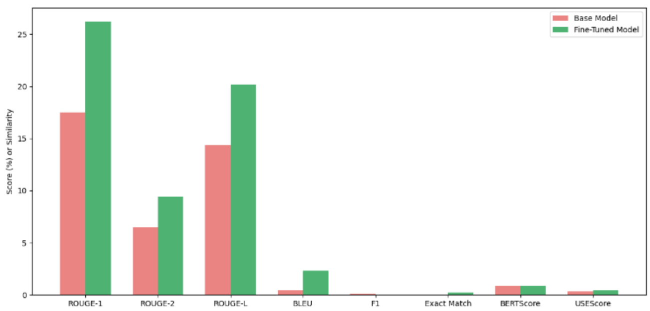

Figure 1: Metric-wise Comparison of Base vs. Fine-Tuned FLAN-T5 on PubMedQA.

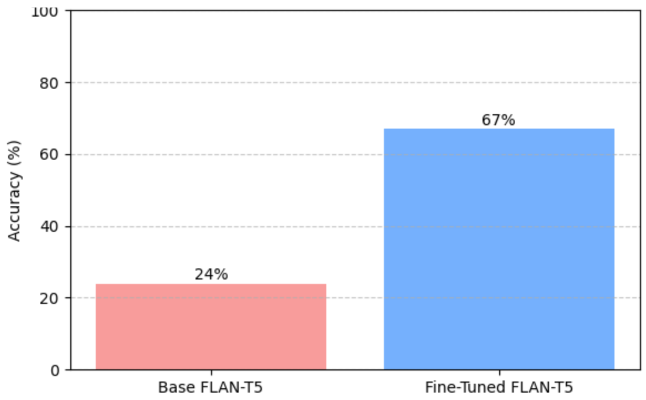

Figure 2: Metric-wise Comparison of Base vs. Fine-Tuned FLAN-T5 on MedMCQA.

METRIC-WISE INTERPRETATION

ROUGE Scores (ROUGE-1, ROUGE-2, and ROUGE-L): These metrics assess how much the generated and reference answers overlap in terms of unigrams, bigrams, and longest common subsequence. [9] The adjusted model demonstrated notable enhancements: ROUGE-1 increased to 26.21% from 17.53%, ROUGE-1 increased to 9.43% from 6.45%. Fluency and structure are measured by ROUGE-L, which rose 5.84%. This implies enhanced answer structure following fine-tuning and better preservation of pertinent content.

BLEU Score: which is frequently used in machine translation, measures n-gram precision. [8] The rise from 0.44% to 2.32% indicates better phrasing and lexical precision, despite the fact that BLEU is less appropriate for generative QA because of answer variability.

F1 Score: captures the equilibrium between recall and precision at the token level. Improvement from 0.1380 to 0.2048 shows that, despite minor form differences, the refined model more closely matches ground truth tokens.

Exact Match (EM): checks to see if the generated response precisely matches the source. Given that PubMedQA responses are free-form and abstractive, it is not surprising that both models received a score of 0.0.

BERTScore (F1): calculates semantic similarity using contextual embeddings. [10] More precise and contextually relevant responses are reflected in the improvement from 0.8462 to 0.8783.

USEScore: calculates the cosine similarity between the reference and predicted answers' Universal Sentence Encoder embeddings [11]. The model's enhanced semantic closeness after fine-tuning is demonstrated by a jump from 0.3191 to 0.4421.

Accuracy: In multiple-choice tests such as MedMCQA, accuracy is the percentage of right answers. The model's accuracy increased from 24% to 67% after it was adjusted. This demonstrates the model's improved comprehension of medical concepts and ability to choose the right response from nearby distractions.

QUALITATIVE ANALYSIS

Manually examined a subset of predictions from the PubMedQA and MedMCQA datasets to supplement the quantitative assessment. Beyond numerical metrics, such as correctness, fluency, and domain relevance, this qualitative analysis aids in understanding the nature of the model's output.

Example Predictions for PubMedQA:

In the biomedical field, the refined FLAN-T5 model shows great potential for producing fluid, contextually relevant responses. Here are a few chosen examples:

PubMedQA Predictions:

Q: Does vitamin D supplementation reduce the risk of respiratory infections?

Context: Vitamin D is known for immune support and may affect respiratory health.

Predicted Answer: Vitamin D is an important component of a healthy immune system and may be a factor in the prevention of respiratory infections.

Q: What are the long-term effects of chemotherapy on cognitive function?

Context: Chemotherapy is a common cancer treatment that may have side effects.

Predicted Answer: Alzheimer's disease is a common side effect of chemotherapy.

Q: Is there a link between gut microbiota and autoimmune diseases?

Context: Gut microbiota influences many aspects of human health and immune function.

Predicted Answer: Gut microbiota plays an important role in the development of autoimmune diseases.

Example Predictions for MedMCQA:

Multiple-choice medical questions are the main focus of this dataset. In this case, the model tries to choose the right answer from the available options. Here are some examples of predictions:

MedMCQA Predictions:

Q: Which vitamin deficiency causes scurvy?

Options: ['Vitamin A', 'Vitamin C', 'Vitamin D', 'Vitamin B12']

Predicted Answer: Vitamin A

Q: The primary function of the sinoatrial node in the heart is to:

Options: ['Pump blood to the lungs', 'Initiate the heartbeat', 'Regulate blood pressure', 'Filter blood']

Predicted Answer: Regulate blood pressure

Q: Which antibiotic is effective against Gram-positive bacteria?

Options: ['Ciprofloxacin', 'Penicillin', 'Metronidazole', 'Amphotericin B']

Predicted Answer: Amphotericin B

Observations:

PubMedQA: The answers are fluent, semantically relevant, and grammatically correct. The necessity for factual consistency checks or post-processing filters is highlighted by the fact that some responses (such as the assertion that "Alzheimer's disease is a common side effect of chemotherapy") are factually inaccurate or unduly generalized. When the context is clear-cut and supported by evidence, the model works well.

MedMCQA: The model often generates a valid answer from the options. Errors indicate a lack of factual foundation or a misinterpretation of biomedical knowledge. Even with adjustments, the multiple-choice format still seems more difficult than open-ended QA, because of distractor options or semantic overlap.

HALLUCINATION AND CONSISTENCY ANALYSIS

In language models, hallucination describes situations in which the model produces information that is confident but factually incorrect. In the medical field, where precision is crucial, this is especially important. [12]

Quantitative Assessments:

Although it can be difficult to detect hallucinations directly, I evaluated factual consistency using the following proxies:

- Exact match: Because PubMedQA is abstractive, it is not surprising that both the base and fine-tuned models received a score of 0.0. The low EM, however, also implies that models frequently depart from the precise wording of the ground truth.
- F1 Score at the Token Level: increased from 0.1380 to 0.2048, suggesting fewer hallucinated tokens and improved alignment with reference answers.
- The BERTScore (F1) increased from 0.8462 to 0.8783, indicating that the predicted and actual answers had better semantic consistency.

Manual Qualitative Insights:

To look for hallucinations, manually reviewed a few outputs:

- PubMedQA: Even when it was wrong, the refined model frequently generated answers that sounded plausible. An example of a typical case of hallucination is the factually unsupported association between Alzheimer's disease and the side effects of chemotherapy.
- MedMCQA: Because of its multiple-choice format, hallucinations are less common. Nonetheless, the base model frequently chose distractor choices (such as "Vitamin A" for scurvy), indicating a conceptual misunderstanding. Better factual grounding was shown by the refined model.

Summary:

Particularly for factual recall and clinical relevance, fine-tuning resulted in fewer hallucinations and increased consistency. The model still occasionally produces confident but inaccurate results, which emphasizes the necessity of domain-specific validation in practical applications.

CONCLUSION AND FUTURE WORK

SUMMARY AND KEY FINDINGS

In this project, I fine-tuned a general-purpose language model (FLAN-T5) on two domain-specific medical datasets: MedMCQA for multiple-choice QA and PubMedQA for generative question answering. Improving factual accuracy, contextual relevance, and semantic alignment in medical question-answering tasks was the main objective.

I noticed noticeable gains in both quantitative metrics and response quality following the fine-tuning process:

- The model demonstrated significant improvements in ROUGE, BLEU, BERTScore, and USEScore on PubMedQA, indicating more fluid and contextually accurate responses.
- Accuracy on MedMCQA rose from 24% to 67%, indicating a better comprehension of clinical knowledge and logic.

CHALLENGES AND SOLUTIONS

Dealing with inconsistent dataset formats was one of biggest obstacles. In order to standardize input-output formats for training, created unique preprocessing pipelines for every dataset. In order to ensure timely model convergence, had to carefully manage batch sizes and limit dataset sizes due to hardware limitations like limited GPU memory and lengthy training times. The need for fine-tuning was further highlighted by the base model's low baseline accuracy, particularly for MedMCQA. After training on pertinent domain data, performance was successfully increased.

LIMITATIONS

Despite the improvements, there were certain limitations in approach:

- Due to computational limitations, only able to fine-tune on a subset of the available data, which might have limited the model's capacity for generalization.
- Automated metrics were the main source of evaluation. These are useful, but they fall short in capturing factual accuracy and clinical correctness.
- Some hallucinations remained, especially in PubMedQA's generative outputs, underscoring the drawbacks of using only pretrained text-based models in delicate fields like medicine.

FUTURE DIRECTIONS

This work establishes a solid foundation for creating trustworthy domain-specific medical question-answering systems. Going forward, future plan is:

- To improve contextual understanding, include multi-modal data, such as clinical images or radiology reports.
- To help the model produce more factual and organized answers, use prompt engineering techniques.
- To lessen hallucinations and increase safety, investigate clinical RLHF (Reinforcement Learning from Human Feedback) with domain experts.

- To ensure that answers are based on credible medical sources, incorporate structured knowledge bases such as UMLS.
- Experiment with cross-domain pretraining, utilizing adjacent domains to enhance knowledge transfer.

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